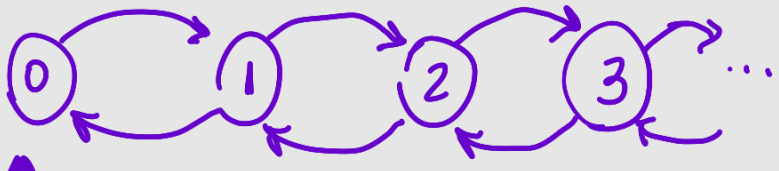
① $\lambda = 3h^{-1}$

$16\,min = \frac{1}{\mu} \rightarrow 0{,}27h = \frac{1}{\mu} \rightarrow \mu = 3{,}75h^{-1}$

a) $\bar{t}_Q = \frac{\lambda}{\mu(\mu-\lambda)} \cong 1h$

b) $\bar{t} = \frac{\lambda}{\mu-\lambda} \cong 1{,}3h$

c)

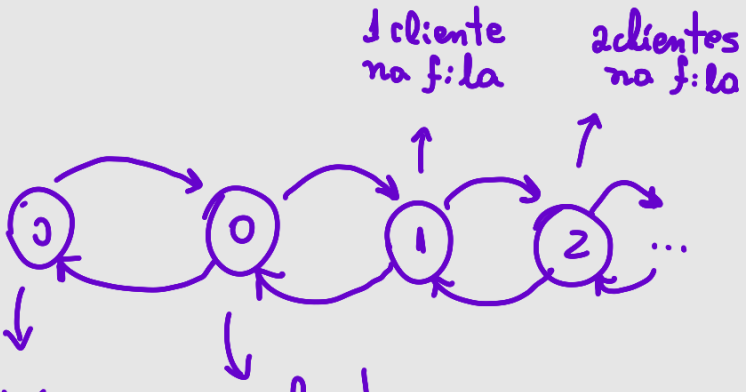


momento que não trabalha $\rightarrow P_0 = 1 - \frac{\lambda}{\mu} = 0{,}2$

② $\lambda = 1\,min^{-1}$

a) $P(N(t) \geq 1) = 1 - P_0$

$P_0 = 1 - \frac{\lambda}{}$



não há espera

1 cliente em atend. → há espera

$1 - P_0 \leq 0{,}05$

$1 - \left(1 - \frac{\lambda}{\mu}\right) \leq 0{,}05$

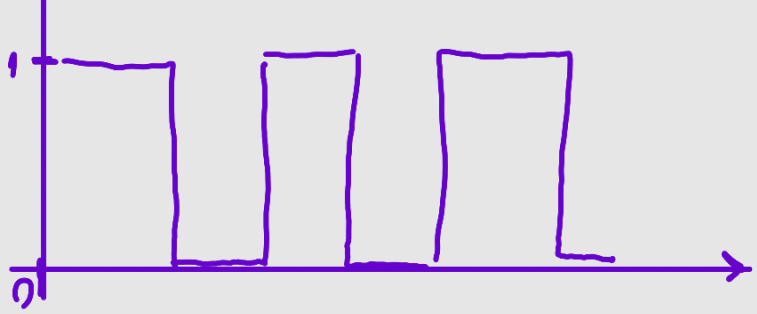
$\frac{1}{\mu} = 0{,}05 = 3s$ por atendimento

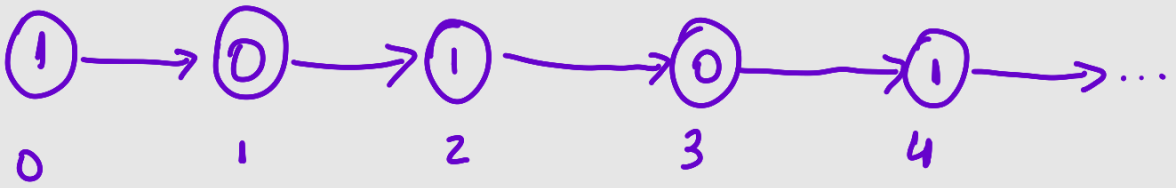
b) $\bar{n}_Q = \frac{\lambda^2}{\mu \cdot (\mu - \lambda)} \rightarrow 2 = \frac{1}{\mu^2 - \mu} \rightarrow \mu^2 - \mu - \frac{1}{2} = 0$

$\mu = 1{,}365 \ min^{-1}$

$\frac{1}{\mu} = 0{,}73 \ min$

(3)





Poisson → Chegada

→ Prob. de ter o valor 1

$$P_0(t) + P_2(t) + P_4(t) + \dots$$

$$= e^{-\lambda t} + \frac{(\lambda t)^2}{2!} \cdot e^{-\lambda t} + \frac{(\lambda t)^4}{4!} \cdot e^{-\lambda t}$$

$$= e^{-\lambda t}\left(1 + \frac{(\lambda t)^2}{2!} + \frac{(\lambda t)^4}{4!} + \dots\right)$$

$$= e^{-\lambda t}\left(\frac{e^{\lambda t} + e^{-\lambda t}}{2}\right) = \frac{1 + e^{-2\lambda t}}{2}$$